SPECTRA: A Traceability Ontology for the 3GPP RAN Standardization Process

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Abstract. Standards-developing organizations produce evolving document corpora whose lifecycle remains under-modeled in the Semantic Web. We present **SPECTRA** (*SPECification TRAcing*), the first OWL ontology for the standardization document lifecycle, instantiated against 3GPP Radio Access Network (RAN) data. SPECTRA is organized around four concerns—contributions, resolutions, specifications, cross-group communication—and grounded in 137 competency questions elicited from practicing standards engineers. The ontology is validated by a cumulative CQ regression protocol across five incremental phases and by cross-WG instantiation: applying the RAN1-designed schema to RAN2–RAN5 required no additional classes, one additional object property, and two ontology-level data properties. Representative CQs with executable Cypher and SPARQL translations, an end-to-end synthetic example, SHACL shapes, and a validation manifest accompany the release.

Keywords: Ontology engineering · Knowledge graph · 3GPP · Competency questions · Telecom standardization

1 Introduction

Standards-developing organizations (SDOs) such as 3GPP, IETF, and IEEE produce evolving document corpora whose lifecycle — from initial contributions through working-group resolutions to normative specification text — shapes which technologies become global infrastructure. The Semantic Web has modeled scholarly publishing, software releases, and provenance, but the SDO standardization *process itself* remains under-modeled despite producing the artifacts that govern interoperability across the telecom, networking, and semiconductor industries. We focus on the largest unmodeled instance: the 3rd Generation Partnership Project (3GPP), the global SDO defining 4G LTE, 5G NR, and the upcoming 6G standards. The Radio Access Network (RAN) Technical Specification Group alone encompasses five Working Groups (RAN1–RAN5), each owning a distinct layer of the radio interface from physical-layer procedures to conformance testing.

3GPP specifications are the normative reference that commercial 5G implementations and conformance tests must align with. Yet the corpus spans

hundreds of normative specifications and evolves through a process that produces hundreds of thousands of Working Group documents. Acquiring even partial working knowledge typically requires substantial domain expertise and prolonged immersion, creating a barrier that compounds along several axes: *cross-WG* (a RAN1 decision propagates to RAN2/3/4 through Liaison Statements few engineers track end-to-end), *cross-organization* (each company has internal context only for its own contributions), and *industry–academia* (researchers see the published specifications but not the meeting-level decisions that shaped them). The traceability path TDoc → Resolution → CR → TS Section → Spec, which captures *why* a clause reads as it does and where it resides, currently exists as implicit institutional knowledge and is typically reconstructed by hand — error-prone and time-consuming.

To the best of our knowledge, no publicly available OWL ontology specifically targets the 3GPP RAN standardization document lifecycle. Recent 3GPP-domain resources — TSpec-LLM [9] for retrieval-augmented access to specification text and *telecom-kg-rel19* [18] for spec-text concept extraction — model the *content* of specifications but are not designed to directly represent the standardization *process* that produced them — e.g., which TDocs led to a clause, which CRs were triggered by a Liaison Statement, or which Working Assumptions were promoted to Agreements.

In this paper, we present **SPECTRA** (*SPECification TRAcing*), a reusable OWL ontology for modeling the types, relationships, and lifecycle of standardization documents, instantiated at scale against 3GPP RAN data. While 3GPP is the validation corpus, the schema's four-concern structure (contributions, resolutions, specifications, cross-group communication) is designed to generalize to sibling SDOs whose document lifecycles share the same shape: contributions are tabled, working-group resolutions decide their fate, change requests amend the normative text, and cross-group communication propagates impact. The contributions are:

1. **The SPECTRA OWL ontology** — a reusable OWL resource for 3GPP RAN standardization documents and their lifecycle, comprising 26 classes and 134 properties (53 object properties, 81 data properties) organized around contributions, resolutions, specifications, and cross-group communication; the ontology reuses Dublin Core, DCTERMS, and FOAF and is licensed CC-BY 4.0 (release details in Section 7).
2. **A competency-question grounding** — SPECTRA's classes, properties, and axioms are derived from 137 competency questions elicited from practicing 3GPP standards engineers at Samsung System LSI; representative CQs with executable Cypher translations and a smaller set of SPARQL examples accompany the public release for inspection and reuse without access to the 3GPP source corpus; the complete suite is retained as internal validation material.
3. **Cross-WG applicability and instantiation evidence** — the schema has been instantiated at realistic scale across all five RAN Working Groups (RAN1–RAN5), and applying the RAN1-designed schema to RAN2–RAN5

required *no additional classes, one additional object property, and two ontology-level data properties* (`summaryId`, `discussionText`; a further 20 loader-local mirror/scaffolding attributes are excluded as instance-data conveniences — see Table 5). The instantiated knowledge graphs and operational parsing pipeline are not redistributed because the underlying TDocs are subject to 3GPP copyright.

4. **Cumulative CQ regression as a quality-assurance mechanism** — alongside established CQ-driven methodologies (LOT [12], SAMOD [11], NeOn [15]), we develop SPECTRA through five incremental phases (25, 34, 45, 15, 18 CQs; 137 total) and re-execute the cumulative suite across phase transitions to detect silently-regressed competencies; three QA cases surfaced were resolved before release.

Sections 2–8 cover related work, 3GPP background, ontology design, the CQ-driven cumulative validation methodology, evaluation, availability/FAIRness, and conclusion respectively.

2 Related Work

2.1 Ontologies for Domain Knowledge Management

Ontologies have been widely adopted as the semantic backbone for domain-specific knowledge management systems. In cultural heritage, the Polifonia Ontology Network [2] provides 15 modular ontologies for musical heritage, developed through eXtreme Design with 361 competency questions from 10 pilots. In the energy domain, the Hydrogen Ontology (HOLY) [1] models the structure of hydrogen-based markets through CQ-driven design. In industry, Cao et al. [4] describe a knowledge graph for the quality management of Bosch products. For technical standards, Grangel-González and Vidal [6] use the Standards Ontology (STO) as the schema for an Industry 4.0 Knowledge Graph covering more than 200 industrial standards, but their focus is on *inter-standard* relationships rather than the *intra-standard* document structure central to our work. In telecommunications, NORIA-O [16] targets anomaly detection and incident management in ICT systems rather than the standardization process itself. The most directly comparable recent effort in the 3GPP domain is *telecom-kg-rel19* [18], a knowledge graph extracted from Release-19 specification *text* for retrieval-augmented LLM reasoning; SPECTRA differs in that it models the standardization *process and document lifecycle* (TDoc, Resolution, CR, LS, TR, Section, etc.) rather than concept structures embedded in spec text. To the best of our knowledge, no publicly available OWL ontology specifically targets the 3GPP RAN standardization document lifecycle and traceability process. Table 1 positions SPECTRA against the closest comparable resources along the axes most relevant to the ISWC Resource Track.

Table 1. Differentiation of SPECTRA against closely related Semantic Web resources along axes most relevant to the ISWC Resource Track, based on each work's published abstract / availability statement. *Process* = explicitly models a domain process/lifecycle. *Multi-hop trace* = supports backward traceability over ≥ 3 class-types. *CQ-grounded* = design driven by an explicit CQ set. *Cross-subdomain validation* = the released schema validated across ≥ 2 sub-domains.

Resource	Domain	Process	Multi-hop trace	CQ-grounded	Cross-subdom. valid.	Real-world deploy.
Polifonia [2]	cultural heritage	limited	no	yes (361 CQs)	yes (15 modules)	multiple stakeholders
HOLY [1]	hydrogen market	no	no	yes (test cases)	no	Atlant-H project
NORIA-O [16]	ICT incidents	yes (incident)	limited	yes (NOC/SOC CQs)	no	telco operator
Grangel & Vidal [6]	I40 standards	no	no	no	no	I40KG
TSpec-LLM [9]	3GPP text	no	no	no	no	LLM benchmark
telecom-kg-rel19 [18]	3GPP Rel-19 text	no	no	no	no	LLM-RAG eval
SPECTRA (ours)	**3GPP RAN process**	**yes**	**yes (5-hop)**	**yes (137 CQs)**	**yes (5 WGs)**	Samsung System LSI

SPECTRA's distinguishing combination is process modelling and multi-hop traceability simultaneously, validated cross-subdomain at instantiation scale; no listed prior resource exhibits all three.

2.2 Ontology Engineering Methodologies

The Ontology 101 guide [10] established a widely-adopted methodology with competency questions (CQs) as the primary requirements mechanism [7]. LOT [12] extends this for industrial settings with FAIR compliance, the NeOn methodology [15] provides scenario-based pathways for building ontology networks (including reuse of existing ontological and non-ontological resources), and SAMOD [11] draws on Test-Driven Development principles. While these methodologies support iterative refinement and, in NeOn's case, modular network development, none explicitly require re-execution of the cumulative CQ suite at each new sub-domain extension, so earlier-phase competencies can be silently regressed. SPECTRA's five-phase cumulative CQ validation (Section 5) uses this protocol as a quality-assurance mechanism for the released resource rather than a replacement for existing methodologies.

In terms of CQ scale, the Polifonia Ontology Network [2] releases a 361-CQ dataset (*PolifoniaCQ*) collected across the network's modules, and Wisniewski et al. [17] analyzed 234 CQs from 5 ontologies and translated them into 131 SPARQL-OWL queries. SPECTRA contributes 137 CQs applied cumulatively to a single cohesive ontology covering five document domains.

Recent LLM-assisted ontology engineering [14] reports near-novice quality on benchmark tasks; systematic comparison for specialized industrial domains is left to future work.

3 3GPP Standardization Background

3.1 Document Types and Lifecycle

3GPP produces six interlinked document types: **TDocs** (Technical Documents — primary contribution unit, e.g., R1-2400001); **Resolutions** (meeting decisions as Agreement/Conclusion/Working Assumption); **CRs** (Change Requests modifying normative text); **TSs** (Technical Specifications, e.g., TS 38.211); **TRs** (Technical Reports — feasibility studies); **LSs** (Liaison Statements — cross-WG communications).

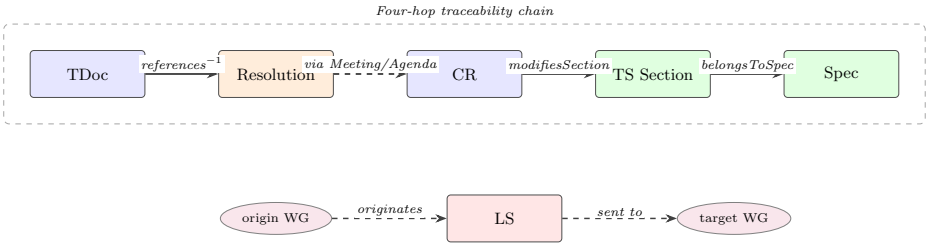

Fig. 1. Document lifecycle in 3GPP RAN. The top row is a reader-facing traceability path from a TDoc contribution to the affected specification: `references` is formally defined Resolution → Tdoc and is drawn in the reading direction, `modifiesSection` links CRs to Sections, and `belongsToSpec` links Sections to Specs. The dashed Resolution–CR step denotes an indirect join through meeting/agenda and CR-pack context rather than a single asserted ontology property. The bottom row summarizes LS-based cross-WG communication via `originatedFrom` and `sentTo`.

3.2 Working Group Structure

The RAN TSG is divided into five WGs, each owning a layer of the radio interface: **RAN1** (PHY, TS 38.211–214), **RAN2** (L2/3, TS 38.331/321), **RAN3** (architecture, TS 38.401/423), **RAN4** (radio/RF, TS 38.101/104), **RAN5** (conformance testing, TS 38.521/533). Cross-WG dependencies are mediated by Liaison Statements; meetings convene every ≈2 months with thousands of TDocs each.

Generalization beyond 3GPP. Although our instantiation targets 3GPP RAN, the four-concern schema (contributions, resolutions, specifications, cross-group communication) maps onto sibling SDOs (Table 2). The lifecycle pattern is the same shape; only the document-type vocabulary differs. We instantiate against 3GPP because it is the largest unmodeled SDO corpus by document volume, but the schema decisions in §4 are deliberately written so that an IETF or IEEE instantiation would only need to substitute the WG-specific document-subtype names rather than redesign the spine.

Table 2. SPECTRA's four-concern schema viewed across three SDOs. The lifecycle shape (contribution → resolution → amendment → normative text) is shared; vocabulary differs.

SDO	Contribution	Resolution	Amendment	Normative text
3GPP (this paper)	TDoc, LS	Agreement / Conclusion / Working Assumption	Change Request (CR)	TS, TR
IETF	Internet-Draft	WG consensus call	RFC errata, bis-drafts	RFC
IEEE 802 / 1722 / etc.	Submission, Comment	Letter Ballot, motion	PAR amendment, sponsor ballot CR	IEEE Std

Cross-SDO empirical validation is left to future work; this table positions SPECTRA's design intent rather than reporting executed mappings.

3.3 Concepts an Ontology Must Capture

Three features shape the ontology: **scale** (10^4–10^5 TDocs per WG; RAN1/2/4 exceed 10^5, RAN3/5 are in 10^4); **document lineage** (TDoc→Resolution→CR→Section→Spec demands first-class entity types and directed links per document); **cross-group dependencies** (LSes as directed artefacts, not free text). SPECTRA's layout (§4) is organized around these three.

4 SPECTRA Ontology Design

4.1 Competency Question Analysis

The design of SPECTRA is driven by 137 competency questions collected across five development phases, in collaboration with domain experts — standardization engineers at Samsung System LSI who implement 3GPP specifications in commercial chipsets. While the full text of every CQ and its executable query is retained as internal validation evidence, an **anonymized index of all 137 CQs** (identifier, phase, category, and schema area exercised) is included in the public release as `cqs/cq_index.md`, so that third parties can verify the breadth of the CQ set and the ontology classes/properties each CQ exercises.

CQs are distributed across the development phases as: P1 TDoc metadata (25, e.g. "What TDocs did Company X submit?"), P2 Meeting resolutions (34, "What agreements on Agenda Item Z?"), P3 TS structure (45, "Which CRs modified section S?"), P4 CR documents (15, "What is the reason for change of CR C?"), P5 Technical reports (18, "Which TRs have impacted TS 38.211?"), totalling 137.

4.2 Core Schema

Based on the CQ analysis, we designed SPECTRA following the Ontology 101 methodology [10]. The ontology is encoded in OWL 2 [8] and comprises 26 classes, 53 object properties, and 81 data properties, organized around four conceptual groups (Figure 2).

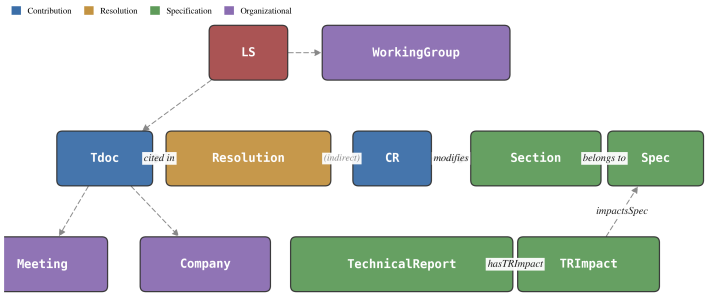

Fig. 2. SPECTRA core schema overview (reader-facing labels in figure; formal OPs below). **Central row**: traceability spine `Tdoc` → `Resolution` → `CR` → `Section` → `Spec`. "*cited in*" = `references` (defined Resolution → Tdoc, drawn reversed); "*modifies*" = `modifiesSection`; "*belongs to*" = `belongsToSpec`. The Resolution → CR edge is drawn lightly because no direct OP is declared; the join is reconstructed via `resolutionBelongsTo` (Resolution → AgendaItem) and `belongsToCRPack` (CR → CRPack) at the same meeting. **Dashed grey arrows** carry secondary relations (LS → Tdoc `rdfs:subClassOf`; LS → WorkingGroup `sentTo`; Tdoc → {Meeting, Company} `presentedAt`/`submittedBy`). The TR → TRImpact → Spec reified chain is shown explicitly. Remaining 26-class members and FOAF reuse are described in §4.

Document: `Tdoc` with four subclasses (`CR`, `LS`, `Summary`, `SessionNotes`); `CR` → `Spec` via `modifies`; `LS` → `WorkingGroup` via `sentTo`/`originatedFrom`. **Resolution**: subclasses `Agreement`, `Conclusion`, `WorkingAssumption` (with `promotedTo` for the Agreement promotion). **Specification**: `Spec` (TS) and `Section` forming a tree via `hasSubSection`/`parentSection` (irreflexive, asymmetric); `TechnicalReport` links to influenced Specs through a reified `TRImpact` so a single TR can affect multiple Specs with per-impact metadata. **Organizational**: `Meeting`, `Company` ($\sqsubseteq$ `foaf:Organization`), `Contact` ($\sqsubseteq$ `foaf:Person`), `WorkItem`, `AgendaItem`, `Release`, `WorkingGroup`.

Key Design Decisions. **D1:** Document types are OWL subclasses (not property values) so that type-specific properties can be declared per subclass. **D2:** 23 object properties carry `owl:FunctionalProperty` for an *at-most-one* contract (e.g., a TDoc is presented at at most one meeting); the loader uses these for ingest-time violation detection and also checks minimum-cardinality, which is not enforced at the OWL level. **D3:** The spine (Figure 2) is realized by three direct OPs (`references`, `modifiesSection`, `belongsToSpec`) and a multi-hop join (Resolution → CR via `resolutionBelongsTo`/`belongsToCRPack`); chain-level OPs are deliberately avoided so each declared property keeps a single, verifiable semantic.

4.3 Reuse of Existing Vocabularies

SPECTRA reuses established Semantic Web vocabularies where they naturally apply: **Dublin Core** [5] (`dc:title`, `description`, `creator`, `date`,

`rights`) and **DCTERMS** (`dcterms:license`) for ontology/instance metadata; **FOAF** [3] (`Person`, `Organization`) as parent classes for `Contact` and `Company`; **OWL 2** axioms (`Class`, `FunctionalProperty`, `InverseFunctionalProperty`, `inverseOf`, `IrreflexiveProperty`, `AsymmetricProperty`); and **RDFS** (`subClassOf`, `label`, `comment`). Domain-specific classes are introduced only where no suitable external term exists.

We considered but did *not* adopt broader vocabularies as the core schema: **ORG** imposes Organization/Unit/Membership structure beyond CQ needs; **SKOS** fits hierarchical term taxonomies, not typed first-class entities; **DCAT/VoID** target dataset description rather than document-process modeling. We do, however, ship an optional **PROV-O alignment** as five `rdfs:subClassOf` axioms appended to the released TTL: `Resolution` $\sqsubseteq$ `prov:Activity`, `Tdoc` $\sqsubseteq$ `prov:Entity`, `Company` $\sqsubseteq$ `prov:Agent` (also `prov:Organization`), `Contact` $\sqsubseteq$ `prov:Agent` (also `prov:Person`). The alignment is declarative (reasoners may exploit it) and does not require importing PROV-O into instance data, so SPECTRA-conformant KGs retain their domain semantics while interoperating with PROV-O-aware tooling.

4.4 Constraints and Axioms

SPECTRA includes 23 `owl:FunctionalProperty` (at-most-one cardinality); 2 `owl:InverseFunctionalProperty` (`hasCR`, `hasTRImpact`); 15 inverse property pairs; 6 `owl:IrreflexiveProperty` covering the section hierarchy, intra-section references, and TR cross-references; and 2 `owl:AsymmetricProperty` on the section hierarchy. These exclude self-loops and 2-cycles in the affected relations. **We deliberately separate semantic commitments encoded in OWL from operational constraints checked by SHACL or the loader**: SHACL shapes (released as `shapes/spectra-core.shacl.ttl`) capture cardinality and class-range constraints reproducibly across triple-store and SHACL-aware tooling, while longer-cycle prevention and minimum-cardinality enforcement remain at the loader level where they require domain-specific recovery semantics.

5 CQ-Driven Cumulative Validation

The cumulative CQ protocol introduced in this section is the quality-assurance mechanism behind the released ontology: each new phase must leave every previously validated competency intact, so that schema refinements are caught at the point of introduction rather than at release.

5.1 Five-Phase Progressive Construction

SPECTRA was developed through five incremental phases, each extending the ontology to cover a new document type while re-executing the full CQ suite accumulated from prior phases. The ontology grew from 11 classes (Phase 1–2, TDoc metadata) to 26 classes (Phase 5, Technical Reports), with the final

v1.0.0 release incorporating refactoring identified through cross-phase integrity verification.

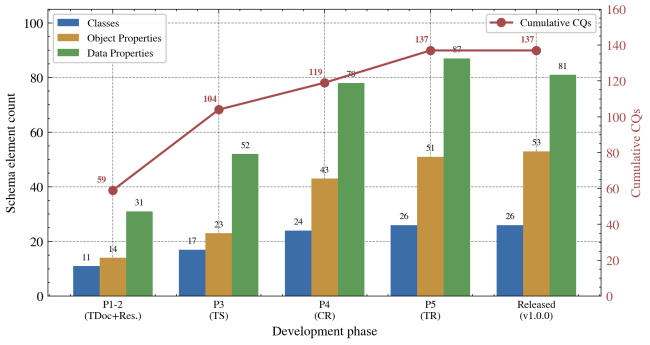

Fig. 3. Schema growth across the five-phase development of SPECTRA, measured from git snapshots at each phase-closing commit. Cumulative CQs (right axis) reach 137 by Phase 5. The data-property drop between Phase 5 and the released version reflects refactoring described in Section 5.

5.2 Quality-Assurance Cases

Rerunning the accumulated CQ suite at each phase transition surfaced concrete schema issues. Three representative cases (traceable via git history) quantify what single-phase validation would have missed: **Case 1** (Phase-2→3, naming inconsistency `MADEAT` vs `MADE_AT`) — 5 Phase-1–2 CQs would have silently regressed; three relationship and three property names harmonized. **Case 2** (Phase-3 audit, meeting-ID referential-integrity gap) — 8 Phase-1–2 CQs broken; three targeted patches, no schema change. **Case 3** (Phase-3→4, three schema refinements: `releaseId` removed as redundant; `hasCR` declared `owl:InverseFunctionalProperty`; base class `Decision` renamed `Resolution`) — 23 Phase-1–2 CQs would have silently broken. Across four transitions (59, 104, 119, 137 cumulative CQs respectively), three of four surfaced regressions that single-phase validation would not have caught.

5.3 Cumulative Regression Verification

At each phase transition, all CQs from Phases 1 through N are re-validated against the schema state introduced by Phase N, executed over an internally instantiated KG conforming to the schema. The 137 CQs accumulated by Phase 5 all return correct, non-empty results (per-CQ status in `validation/cq_results.json`); after the QA cases above were resolved, our internal validation records did not surface further regressions at subsequent phase transitions (an internal observation, since the instantiated KG is not part of the public release).

6 Evaluation

We organise the evaluation around five dimensions appropriate to a copyright-aware Resource Track artifact: (E1) requirements coverage by CQs, (E2) structural quality of the schema, (E3) scalability evidence at instantiation, (E4) cross-WG generality of the schema, and (E5) reusability evidence via public artifacts. The rationale of the copyright-aware release design itself is detailed in Section 7.

6.1 E1 — Requirements coverage by CQs

All 137 CQs (137/137 PASS; per-phase 25/34/45/15/18; per-CQ status in `validation/cq_results.json`) return non-empty results against the internally instantiated KG; correctness was confirmed by the engineers who elicited them (a threat to validity acknowledged in §8). The 137 CQs collectively exercise 20/26 classes and 31/53 OPs (`validation/cq_coverage.json`; phase distribution in `diagrams/cq_distribution.png`). Unexercised by the numbered CQs are auxiliary content classes (Chart/Figure/Table/Summary/SessionNotes) and OPs reached only via inverse paths or join patterns (e.g., `modifiesSection`, `promotedTo`); the released `MULTI_HOP_traceability` example traverses both. `LS` is uncovered under the numbered CQs but central to the cross-WG queries (§6.4).

6.2 E2 — Structural quality

We report structural metrics of the released ontology (Table 3) and the findings of an automated pitfall scan.

Table 3. Ontology structural metrics.

Metric	Value
Classes	26
Object Properties	53
Data Properties	81
Functional Properties	23
Inverse Functional Properties	2
Inverse Property Pairs	15
Irreflexive Properties	6
Asymmetric Properties	2
Reused vocabulary terms	8 (5 DC + 1 DCTERMS + 2 FOAF)

OOPS! [13] reports 0 Critical, 2 Important, 2 Minor pitfalls (full output: `validation/oops_summary.json`). The Important findings are expected design consequences: **P34** (2 affected) flags `foaf:Person`/`foaf:Organization` which we reuse without redeclaring per FOAF best practice; **P10** (1 affected) flags

one missing `owl:disjointWith` between Tdoc subclasses, deferred because released CQs do not rely on disjointness. The Minor pitfalls — P08 (annotations on reused FOAF, 2 affected) and P13 (inverses declared only where bidirectionally queried, 23 affected) — reflect the same reuse-and-parsimony posture. The release also ships a PyLODE-rendered HTML and a SHACL shapes file (`spectra-core.shacl.ttl`) validated against the bundled synthetic instantiation via pySHACL.

6.3 E3 — Scalability evidence at instantiation

To provide internal evidence that the schema tolerates realistic document volumes, we instantiated SPECTRA against RAN1 data. The graph contains 123,677 Technical Documents from 60 meetings, together with all associated Resolutions, Change Requests, Liaison Statements, Specifications, and Technical Reports (Table 4). Referential integrity for the `submittedBy` relation, the most heavily relied on by the competency questions, is 99.36% (1,000 missing links out of 155,525 source TDoc-list records, of which 123,677 became `Tdoc` nodes; the remainder are revisions and deferred entries); the residual gap is traceable to missing source metadata in the original 3GPP TDoc list rather than to schema inadequacy. This instantiation is maintained as internal validation evidence; per Section 7 it is not part of the public release because the underlying TDocs are subject to 3GPP copyright.

Table 4. Per-class instance counts in the RAN1 instantiation of SPECTRA (internal validation evidence).

Class	#	Class	#	Class	#
Tdoc	123,677	Resolution	24,710	Section	1,003
CR	10,715	Agreement	21,058	TSTable	680
LS	6,343	Conclusion	2,703	TSFigure	13
Summary	3,468	WorkingAssump.	949	CRPack	501
SessionNotes	678	Meeting	60	TechnicalReport	26
Spec	83	Company	222	TRImpact	29
Release	13	Contact	1,000	Figure	120
WorkingGroup	116	WorkItem	423	Table	1,441
AgendaItem	6,439			Chart	1

6.4 E4 — Cross-WG generality of the schema

A schema designed for one WG can fail silently when applied to a sibling WG with different document conventions. We applied SPECTRA — originally designed from RAN1 CQs — to RAN2–RAN5 without redesign and measured a stricter quantity than label overlap: how many *additional* schema elements each WG needed beyond what the RAN1-designed ontology already defined. Table 5 reports the schema-level diff computed from each WG's instantiated KG.

Table 5. Schema diff of RAN2–RAN5 instantiations against the RAN1-designed SPECTRA schema. "WG-specific extensions" counts schema elements present in a non-RAN1 WG graph but not declared in the released ontology (union over RAN2–RAN5).

Schema element	In released SPECTRA	WG-specific extensions	Extension rate
Classes	26	0	0.0%
Object Properties	53	1	1.9%
Data Properties	81	2 †	2.5%

† excluding 20 loader-local properties (17 underscore-prefixed mirror slots and 3 internal scaffolding properties).

The 26 classes cover every document-type distinction the per-WG loaders produced (**zero** additional classes); only one additional object property (`REFERENCES_SPEC`) appears across all four non-RAN1 WGs. The 22 apparent data-property extensions decompose into 20 loader-local attributes (17 underscore-prefixed mirror slots, e.g., `_submittedBy`; 3 scaffolding properties: `id`, `release`, `for_` — none of which represent ontology-level commitments) and two genuinely new data properties (`discussionText`, `summaryId`). **Per-WG ontology-level extensions** (classes / OPs / genuine DPs): RAN2 *0/1/0*, RAN3 *0/1/1* (`summaryId`), RAN4 *0/1/0*, RAN5 *0/1/1* (`discussionText`); the single OP (`REFERENCES_SPEC`) is shared across all four. The total additional schema surface across four sibling WGs is therefore one OP and two DPs on top of the 160 elements already declared — evidence that the RAN1 schema is stable under cross-WG application within the RAN TSG rather than a universal transferability claim.

Conversely, 6 of the 26 RAN1 classes are absent from at least one sibling WG's instantiation. **Per-WG class coverage** (released in `validation/`): RAN1 26/26 (100%), RAN2 22/26 (84.6%), RAN3 23/26 (88.5%), RAN4 24/26 (92.3%), RAN5 20/26 (76.9%) (RAN5, scoped to conformance testing, omits Agreement, WorkingAssumption, Summary, SessionNotes, Chart, and Figure). These absences are expected and not counted as failures: they indicate that a non-RAN1 WG does not need to populate the full RAN1 ontology to benefit from reuse.

6.5 E5 — Reusability evidence via public artifacts

To illustrate what the ontology is for, we list a curated set of competency questions that SPECTRA supports against an instantiated knowledge graph; each question is paired with the schema fragment it exercises and the role it plays in the standardization workflow.

- **S1. Multi-hop traceability.** "Which TDoc motivated the current text of TS 38.214 §5.1.3?" Starts at the Section, traverses CRs that modify it (reverse traversal over `modifiesSection`), joins through the Meeting at which the CR was presented to find the Resolution made there, then follows

`references` to the originating contribution. *Schema fragment:* `Section, CR, Meeting, Resolution, Tdoc; MODIFIES_SECTION, PRESENTED_AT, MADE_AT, REFERENCES`. *Why it matters:* reconstructs the contribution that justifies a clause's current wording — otherwise hours of manual archive search.

- **S2. Cross-WG impact via Liaison Statements.** "Which RAN2 contributions in meeting M were triggered by RAN1 LSes?" Uses `originatedFrom/sentTo/replyIn`. *On deployed KGs:* RAN1 sends 3,644 LSes to RAN2 (peak recipient; RAN4 1,457, RAN3 380); RAN2 receives 1,450 RAN1-originated LSes (peak meeting RAN2#118-e).
- **S3. Release-scoped CR analytics.** "CRs against Release 18 by Company X" combines `targetRelease`, `modifies`, `submittedBy`. *On deployed KGs:* TS 38.300 is jointly modified by 2,039 RAN2 CRs and 270 RAN3 CRs, surfaced via `modifies` without per-WG customisation.
- **S4. Working Assumption promotion lineage.** "Show all WAs eventually promoted to Agreements, and the meeting span." Walks $\texttt{WorkingAssumption} \xrightarrow{\texttt{promotedTo}} \texttt{Agreement}$ joined on `madeAt`. *Why it matters:* surfaces which speculative decisions hardened into binding agreements — not answered by any single per-WG report.
- **S5. TR-to-TS impact propagation.** "Which TS sections are affected by TR 38.913 study conclusions?" Walks $\texttt{TR} \xrightarrow{\texttt{hasTRImpact}} \texttt{TRImpact} \xrightarrow{\texttt{impactsSection}} \texttt{Section}$, exercising the reified `TRImpact` class for one-TR-to-many-TS impact with per-impact metadata.

The executable form (15 representative Cypher + 6 SPARQL queries, plus an end-to-end synthetic example bundling data, queries, and expected output) is shipped under `queries/` and `examples/end_to_end/` so reusers can verify their toolchain before instantiating SPECTRA on their own data. The full 137-CQ executable suite is retained as internal validation material (§5). SPECTRA also underwrites a standards-engineering knowledge infrastructure at the eliciting organisation.

7 Availability, Sustainability, and FAIRness

We follow a *copyright-aware FAIR release pattern*: schema, validation evidence, and reusable query examples are released openly, while the instantiated KG and parsing pipeline remain internal due to 3GPP TDoc redistribution restrictions. A `validation/` directory of JSON evidence plus `validation_manifest.md` maps every quantitative claim in this paper to its supporting file, so reviewers can inspect each cited number directly.

Parsing pipeline (system overview). A SPECTRA-conformant KG is produced in five stages: (i) 3GPP portal scraping (per-WG TDoc lists + document files), (ii) deterministic Tdoc/Resolution metadata parsers → JSON-LD, (iii) LLM-assisted extraction (OpenRouter Gemini-2.5-flash)

for CR `reasonForChange`/`summaryOfChange` and TR scope/conclusions, (iv) SHACL validation, (v) bulk Neo4j load via parallel Cypher `CREATE` (`ProcessPoolExecutor`, checkpointed resume). Throughput: ≈ 2,000 TDocs/min metadata-only, ≈ 50 CRs/min LLM-assisted. The pipeline source is internal; the recipe is sufficient for a third party to build an equivalent loader.

Publicly released (CC-BY 4.0).

- **SPECTRA ontology** (OWL 2 Turtle); the ontology declares `https://w3id.org/spectra` as its IRI, to be registered with the W3C Permanent Identifier service before the camera-ready version. **PyLODE HTML documentation** and a **SHACL shapes file** for core cardinality/range constraints.
- **README** (scope, version, license) and machine-readable metadata (`CITATION.cff`, `codemeta.json`); a schema diagram.
- **Representative competency questions** (curated subset of the 137-CQ suite) with 15 **executable Cypher translations** and 6 **SPARQL examples**, plus a **synthetic instantiation snippet**.
- **Validation evidence** (`validation/`): JSON files for structural metrics, OOPS! pitfall scan, cross-WG schema diff, and cross-WG use-evidence counts, each linked to a paper claim by `validation_manifest.md`.

Not publicly released.

- The complete 137-CQ validation suite and the full set of executable Cypher queries used for cumulative regression testing (retained as internal validation material; representative subsets are released as noted above).
- The instantiated 3GPP RAN knowledge graphs (per-WG instantiations at realistic scale; not redistributable because the underlying TDocs are subject to 3GPP copyright; the original TDocs remain publicly accessible via the 3GPP portal at `https://www.3gpp.org`).
- The operational parsing and knowledge-graph population pipeline (internal project code; not released).

FAIR alignment. **F:** a persistent Zenodo DOI will be minted from the tagged GitHub release upon camera-ready publication. **A:** the OWL file and documentation are downloadable without authentication. **I:** OWL 2 Turtle (loadable by RDFLib in our pipeline; supported by standard tooling such as Protégé and Jena); reuses DC/DCTERMS and FOAF. **R:** CC-BY 4.0, versioned, with README, representative queries, and the validation manifest above.

Sustainability. SPECTRA serves as the semantic layer of a standards-engineering knowledge infrastructure at the eliciting organisation; incremental updates per 3GPP meeting cycle are produced internally, validated against the cumulative CQ protocol before release. The v1.0.0 release remains usable as a CC-BY 4.0 self-contained snapshot for community continuation should internal maintenance lapse.

8 Conclusion

We presented **SPECTRA** as a reusable OWL ontology resource for 3GPP RAN standardization (CC-BY 4.0, Zenodo DOI upon publication), grounded in 137 CQs validated by a five-phase cumulative protocol; the released schema is shown to be cross-WG stable across RAN1–RAN5 with only one additional OP and two genuine DPs needed for RAN2–RAN5. The instantiated KGs and parsing pipeline are retained as internal validation evidence (3GPP copyright, Section 7).

Limitations and future work. Scope is the RAN TSG; extending to TSG SA/CT would add classes and relationships. The released queries are a representative subset of the full 137-CQ executable suite, retained internally with the per-WG KGs and parsing pipeline (3GPP copyright); no public SPARQL endpoint is provided. Adoption evidence is currently confined to the authors' organisation, and CQ correctness was verified by the eliciting engineers themselves. We plan to extend to TSG SA/CT, investigate LLM-assisted CQ generation, examine other standards bodies (IETF, IEEE), and engage the Semantic Web community for independent reuse evidence.

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Resource Availability Statement: The SPECTRA ontology, SHACL shapes, PyLODE HTML documentation, anonymized 137-CQ index, 15 Cypher and 6 SPARQL representative queries, end-to-end synthetic example, and `validation/` directory (with `validation_manifest.md` mapping every quantitative claim to its supporting JSON) are publicly released under CC-BY 4.0 at `https://github.com/spectra-ontology/spec-trace`; a Zenodo DOI minted from the tagged release will replace this URL upon camera-ready publication. The full 137-CQ executable suite, per-WG KGs, and parsing pipeline are retained internally because the underlying 3GPP TDocs are subject to copyright; the TDocs remain publicly accessible via `https://www.3gpp.org`.

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